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SLOT - D

27

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I S.Nirmal (192511249) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Nirmal

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	1	1	2	2
Solution Process	25%	1	2	1	1	1
Accuracy of Calculation	20%	1	1	2	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 35 /40

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer 1: Subnetting the 192.168.20.0/24 Block for Labs A, B and C

1. Problem Understanding

The university has been given one Class C private block, **192.168.20.0/24**, which provides 256 addresses in total. It must be divided into three subnets of different sizes using fixed masks: **Lab A → /26**, **Lab B → /27** and **Lab C → /28**. For every lab the following must be found: the network address, the broadcast address, the valid host range and the number of usable hosts. Because the masks are different, this is a variable-length subnet mask (VLSM) allocation.

2. Method Selection

The block-size (magic number) method is used, because it is quick, avoids long binary conversions and is easy to verify. The formulas are:

$$\text{Host bits } (h) = 32 - \text{prefix length}$$

$$\text{Block size} = 2^h \qquad \text{Usable hosts} = 2^h - 2$$

$$\text{Network address} = \text{first address of the block} \qquad \text{Broadcast address} = \text{network address} + \text{block size} - 1$$

The largest subnet is placed first so that every subnet starts on a correct boundary and no two subnets overlap. Figure 1 shows the procedure that is followed in Answers 1 to 4.

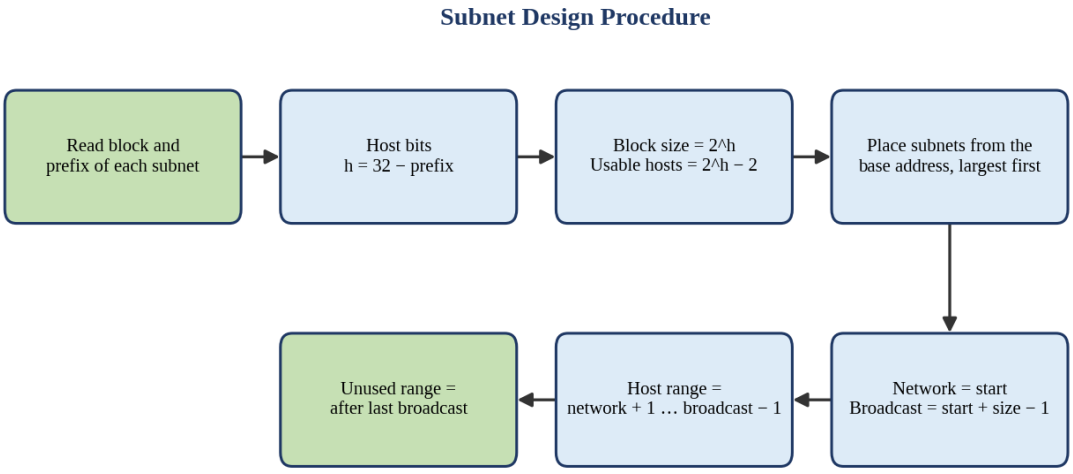


Figure 1: Flowchart of the subnetting procedure used in the answers

3. Solution Process

Lab	Prefix	Subnet mask	Host bits	Block size	Usable hosts ($2^h - 2$)
Lab A	/26	255.255.255.192	6	64	$2^6 - 2 = 62$
Lab B	/27	255.255.255.224	5	32	$2^5 - 2 = 30$
Lab C	/28	255.255.255.240	4	16	$2^4 - 2 = 14$

Step 1 – Lab A (/26). The first block starts at the base address 192.168.20.0, so the network address is 192.168.20.0. The block size is 64, so the broadcast address is 0 + 64 − 1 = 63 → 192.168.20.63. The hosts run from .1 to .62.

Step 2 – Lab B (/27). The next free address after .63 is .64. The block size is 32, so the broadcast is 64 + 32 − 1 = 95. A /27 network must start on a multiple of 32, and 64 = 2 × 32, so the boundary is valid. The hosts run from .65 to .94.

Step 3 – Lab C (/28). The next free address after .95 is .96. The block size is 16, so the broadcast is 96 + 16 − 1 = 111. The boundary is valid because 96 = 6 × 16. The hosts run from .97 to .110.

Subnet	Network address	Subnet mask	Valid host range	Broadcast	Usable hosts
Lab A	192.168.20.0/26	255.255.255.192	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	192.168.20.64/27	255.255.255.224	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	192.168.20.96/28	255.255.255.240	192.168.20.97 – 192.168.20.110	192.168.20.111	14

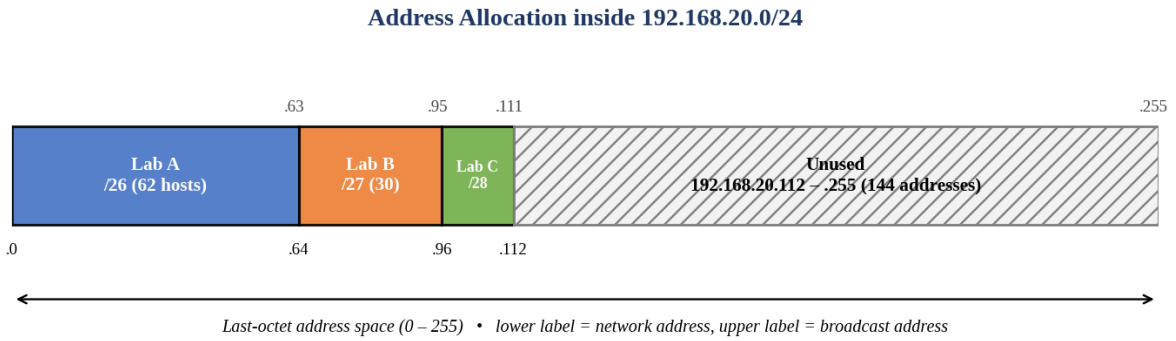


Figure 2: Allocation of the three lab subnets within the 192.168.20.0/24 block

Step 4 – Remaining space. The last address used is 192.168.20.111, so the free range is 192.168.20.112 – 192.168.20.255, which is 256 − 112 = 144 addresses. Total usable hosts over the three labs = 62 + 30 + 14 = **106**.

Result and Interpretation: Lab A can support 62 devices, Lab B 30 devices and Lab C 14 devices, and each subnet is isolated from the others by its own network and broadcast address. The 144 free addresses (192.168.20.112/28 onwards) are kept as a reserve, so new labs can be added without renumbering the existing ones.

Answer 2: Analysing System 1 (192.168.1.130) and System 2 (192.168.1.190)

1. Problem Understanding

Two systems use the same subnet mask 255.255.255.192. System 1 has the IP address 192.168.1.130 and System 2 has 192.168.1.190. The task is to find the subnet range, network address, broadcast address and valid host range for each system, to check that each IP address lies inside its own subnet and to count the usable hosts.

2. Method Selection

The mask 255.255.255.192 equals /26, because 192 = 11000000 uses two bits of the last octet. The **bitwise AND** of the IP address and the mask gives the network address, and setting all host bits to 1 gives the broadcast address. The block size is 256 − 192 = 64, so subnets in the last octet start at 0, 64, 128 and 192.

3. Solution Process

Step 1 – Binary AND for System 1.

```
Last octet of IP   130 = 1 0 0 0 0 0 1 0
Last octet of mask 192 = 1 1 0 0 0 0 0 0
---------------------------------------
AND result         128 = 1 0 0 0 0 0 0 0   -> network address 192.168.1.128
Host bits set to 1 191 = 1 0 1 1 1 1 1 1   -> broadcast address 192.168.1.191
```

Step 2 – Binary AND for System 2.

```
Last octet of IP   190 = 1 0 1 1 1 1 1 0
Last octet of mask 192 = 1 1 0 0 0 0 0 0
---------------------------------------
AND result         128 = 1 0 0 0 0 0 0 0   -> network address 192.168.1.128
Host bits set to 1 191 = 1 0 1 1 1 1 1 1   -> broadcast address 192.168.1.191
```

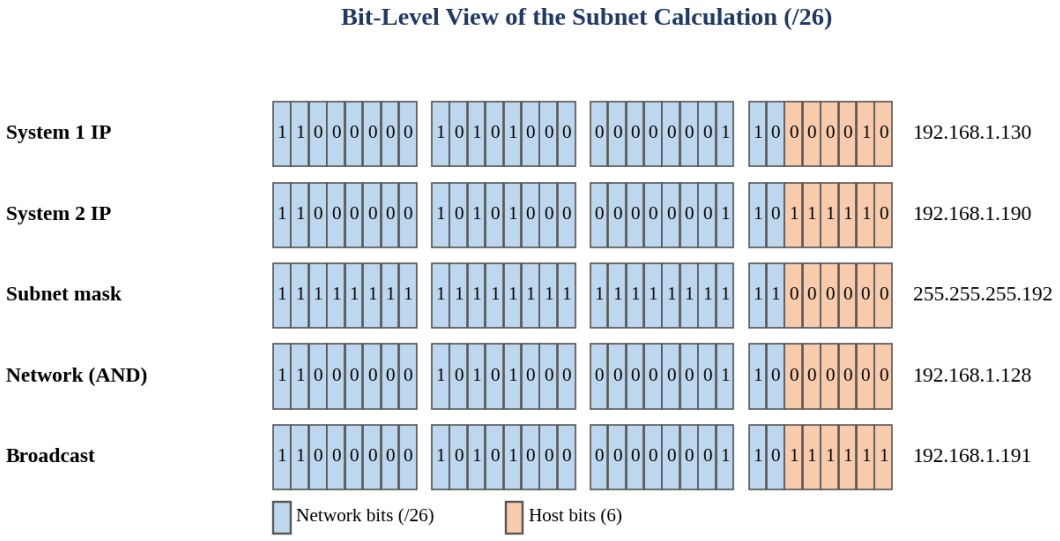


Figure 3: Network and host bits for both systems; the first 26 bits are identical

Step 3 – Range check. The subnet is 192.168.1.128/26 with the broadcast at 192.168.1.191 and the valid host range 192.168.1.129 – 192.168.1.190. System 1: 128 ≤ 130 ≤ 191, so it lies inside the subnet. System 2: 128 ≤ 190 ≤ 191, so it also lies inside the subnet.

Parameter	System 1	System 2
IP address / mask	192.168.1.130 / 255.255.255.192	192.168.1.190 / 255.255.255.192
Subnet range	192.168.1.128 – 192.168.1.191	192.168.1.128 – 192.168.1.191
Network address	192.168.1.128	192.168.1.128
Broadcast address	192.168.1.191	192.168.1.191
Valid host range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
Is the IP inside the subnet?	Yes (130)	Yes (190, the last usable host)
Usable hosts	$2^6 - 2 = 62$	$2^6 - 2 = 62$

Result and Interpretation: Both systems belong to the **same subnet**, 192.168.1.128/26, so they can exchange traffic directly at Layer 2 without a router. System 2 uses 192.168.1.190, the last assignable host address; it is valid, but 192.168.1.191 must never be assigned because it is the broadcast address. The subnet offers 62 usable host addresses.

Answer 3: Allocating Department Subnets from 172.16.0.0/16

1. Problem Understanding

The organization owns the private block **172.16.0.0/16** (65,536 addresses). The masks are fixed: HR → /25, Sales → /26, IT → /27 and Admin → /28. Suitable non-overlapping subnet addresses must be assigned, each subnet must be described completely, and the unused part of the /16 block must be stated.

2. Method Selection

VLSM allocation in **descending order of size** is selected. Placing the largest subnet first keeps every network address aligned to a multiple of its block size and leaves one continuous free block at the end. The /16 prefix fixes the first two octets (172.16), so all four subnets fit inside the first /24 (172.16.0.0/24) because their combined size is only 128 + 64 + 32 + 16 = 240 addresses.

3. Solution Process

Department	Prefix	Mask	Host bits	Block size	Usable hosts
HR	/25	255.255.255.128	7	128	$2^7 - 2 = 126$
Sales	/26	255.255.255.192	6	64	$2^6 - 2 = 62$
IT	/27	255.255.255.224	5	32	$2^5 - 2 = 30$
Admin	/28	255.255.255.240	4	16	$2^4 - 2 = 14$

Step 1 – HR. Start at 172.16.0.0; the broadcast is 0 + 128 − 1 = 127. **Step 2 – Sales.** The next free address is .128 (a multiple of 64); the broadcast is 128 + 64 − 1 = 191. **Step 3 – IT.** The next free address is .192 (a multiple of 32); the broadcast is 192 + 32 − 1 = 223. **Step 4 – Admin.** The next free address is .224 (a multiple of 16); the broadcast is 224 + 16 − 1 = 239.

Subnet	Network address	Subnet mask	Valid host range	Broadcast	Usable hosts
HR	172.16.0.0/25	255.255.255.128	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	172.16.0.128/26	255.255.255.192	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	172.16.0.192/27	255.255.255.224	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	172.16.0.224/28	255.255.255.240	172.16.0.225 – 172.16.0.238	172.16.0.239	14

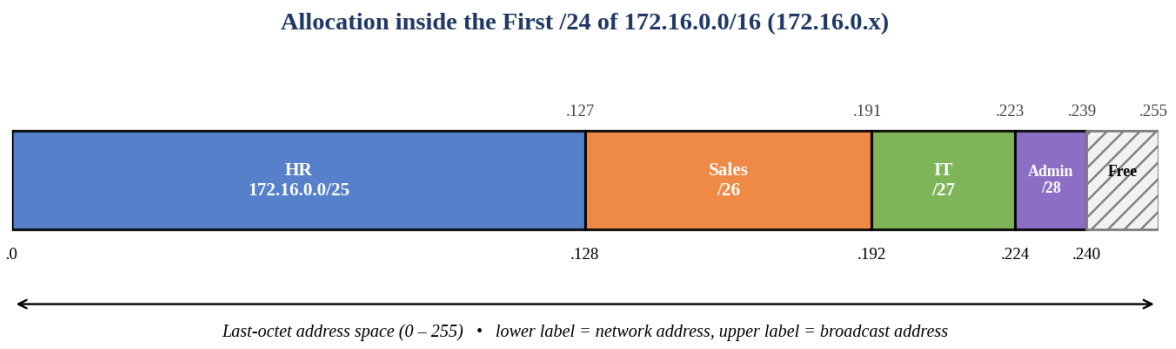


Figure 4: Department subnets placed from the base address of 172.16.0.0/16

Step 5 – Unused range. The last address in use is 172.16.0.239, so the unused space starts at **172.16.0.240** and runs to **172.16.255.255**. Addresses used = 128 + 64 + 32 + 16 = 240, so unused addresses = 65,536 − 240 = **65,296**. Of these, 16 addresses (172.16.0.240/28) are left at the end of the first /24, and the whole range 172.16.1.0 – 172.16.255.255 (255 further /24 networks) is untouched.

Result and Interpretation: HR, Sales, IT and Admin can host 126, 62, 30 and 14 devices respectively, a total of 232 usable hosts, without any overlap. Because only 240 of the 65,536 addresses are used, more than 99.6% of the block is available for new departments, servers and WAN links.

Answer 4: Subnetting 172.16.10.0/24 with a /27 Mask

1. Problem Understanding

The company uses 172.16.10.0/24 and borrows bits to create /27 subnets. The number of subnets and the usable hosts per subnet must be found, together with the subnet of 172.16.10.78 (its network and broadcast address) and a decision on whether 172.16.10.95 belongs to the same subnet.

2. Method Selection

$$\text{Subnets created} = 2^{s}, \text{ where } s = \text{borrowed bits} = 27 - 24 = 3$$

$$\text{Usable hosts per subnet} = 2^{h} - 2, \text{ where } h = 32 - 27 = 5$$

To locate a host, the block size (256 − 224 = 32) is used: the network address is the largest multiple of 32 that is less than or equal to the host value.

3. Solution Process

Step 1 – Number of subnets and hosts. Borrowed bits s = 3, so subnets = 2^3 = **8**. Host bits h = 5, so usable hosts = 2^5 − 2 = **30** per subnet. The mask is 255.255.255.224.

Subnet	Network address	Valid host range	Broadcast
1	172.16.10.0	172.16.10.1 – 172.16.10.30	172.16.10.31
2	172.16.10.32	172.16.10.33 – 172.16.10.62	172.16.10.63
3	172.16.10.64	172.16.10.65 – 172.16.10.94	172.16.10.95
4	172.16.10.96	172.16.10.97 – 172.16.10.126	172.16.10.127
5	172.16.10.128	172.16.10.129 – 172.16.10.158	172.16.10.159
6	172.16.10.160	172.16.10.161 – 172.16.10.190	172.16.10.191
7	172.16.10.192	172.16.10.193 – 172.16.10.222	172.16.10.223
8	172.16.10.224	172.16.10.225 – 172.16.10.254	172.16.10.255

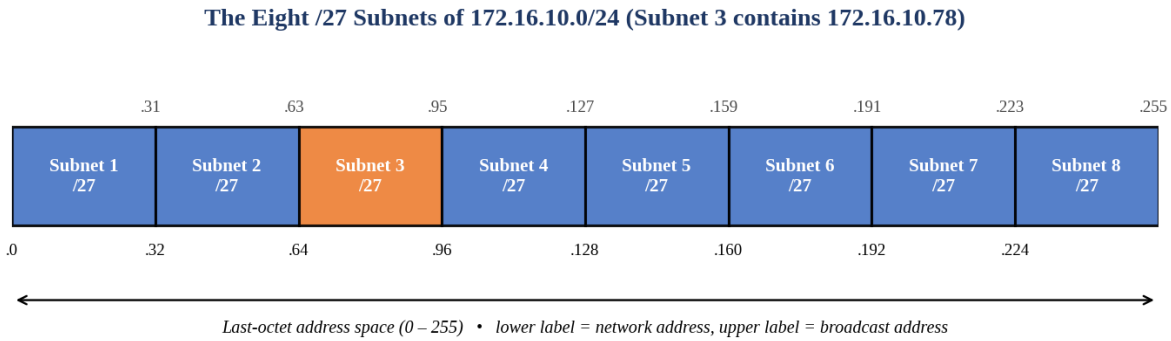


Figure 5: Subnet map; the highlighted block (Subnet 3) holds the host 172.16.10.78

Step 2 – Subnet of 172.16.10.78. 78 ÷ 32 = 2.43, so the integer part is 2 and the network address is 2 × 32 = 64. The same result follows from the binary AND:

```
78  = 0 1 0 0 1 1 1 0
224 = 1 1 1 0 0 0 0 0
---------------
AND = 0 1 0 0 0 0 0 0 = 64   -> network 172.16.10.64
Host bits all 1 = 0 1 0 1 1 1 1 1 = 95   -> broadcast 172.16.10.95
```

Hence 172.16.10.78 belongs to the subnet **172.16.10.64/27**, whose network address is 172.16.10.64, broadcast address is 172.16.10.95 and valid host range is 172.16.10.65 – 172.16.10.94.

Step 3 – Is 172.16.10.95 in the same subnet? The subnet spans 172.16.10.64 – 172.16.10.95, and 64 ≤ 95 ≤ 95, so the address lies inside the range of the same subnet. However, all five host bits of 95 are 1, so it is the **broadcast address** of the subnet, and it cannot be assigned to any device.

Result and Interpretation: The /27 mask creates 8 subnets with 30 usable hosts each. The host 172.16.10.78 sits in 172.16.10.64/27. The address 172.16.10.95 is in the same range but is the broadcast address, so it is not a valid host address; the last assignable host is 172.16.10.94.

Answer 5: IPv6 Address Compression, Expansion, Prefix and Host Capacity

1. Problem Understanding

The IPv6 address **2001:0db8:0000:0000:0000:ff00:0042:8329** is given. It must be compressed to its shortest form, expanded back to full notation, its /64 network prefix identified and the number of host addresses available in the network calculated.

2. Method Selection

IPv6 has 128 bits written as eight 16-bit groups of hexadecimal digits separated by colons. Two rules govern shortening: (1) leading zeros inside a group may be dropped; (2) **one** run of consecutive all-zero groups (the longest, or the leftmost if equal) may be replaced by a double colon "::". The prefix is the first /64 bits (the first four groups), and the host capacity is $2^{128-\text{prefix}}$.

3. Solution Process

Step 1 – Compression.

```
Full form             : 2001:0db8:0000:0000:0000:ff00:0042:8329
Rule 1 (drop leading 0): 2001:db8:0:0:0:ff00:42:8329
Rule 2 (:: for 0:0:0)  : 2001:db8::ff00:42:8329     <- shortest form
```

Step 2 – Expansion. The compressed address shows five explicit groups (2001, db8, ff00, 42, 8329), so "::" stands for 8 − 5 = 3 missing groups of zeros. Each group is then padded to four digits:

```
2001:db8::ff00:42:8329  ->  2001:0db8:0000:0000:0000:ff00:0042:8329
```

Step 3 – Network prefix (/64). The first 64 bits are the first four groups: 2001:0db8:0000:0000, written as **2001:db8::/64** (2001:db8:0:0::/64). The remaining 64 bits, 0000:ff00:0042:8329 (written ::ff00:42:8329), form the interface identifier.

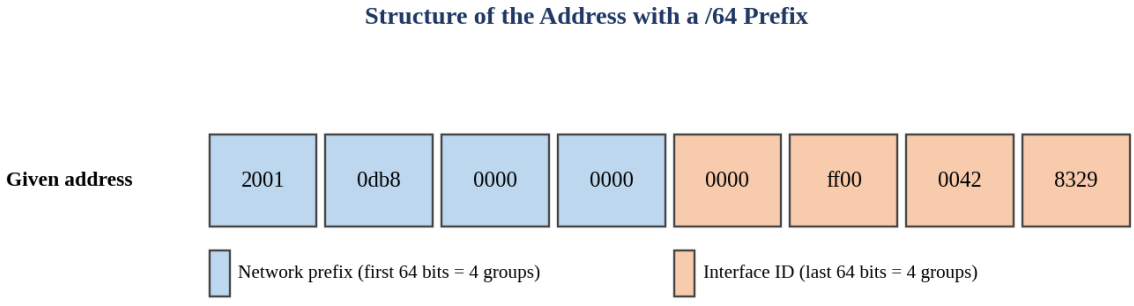


Figure 6: Division of the IPv6 address into network prefix and interface identifier

Step 4 – Host capacity. Host bits = 128 − 64 = 64, so the number of addresses = 2^{64}.

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616 \approx 1.84 \times 10^{19} \text{ addresses}$$

IPv6 has no broadcast address, so the "−2" used in IPv4 is not applied. In practice the all-zeros interface identifier is reserved as the subnet-router anycast address, so about $2^{64} - 1$ addresses can be assigned to hosts.

Result and Interpretation: The address compresses to 2001:db8::ff00:42:8329, expands back without loss, and belongs to the network 2001:db8::/64. This single subnet can contain 18,446,744,073,709,551,616 addresses, far more than the 254 hosts of an IPv4 /24, which removes the need for address-conserving techniques such as NAT inside the subnet.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation